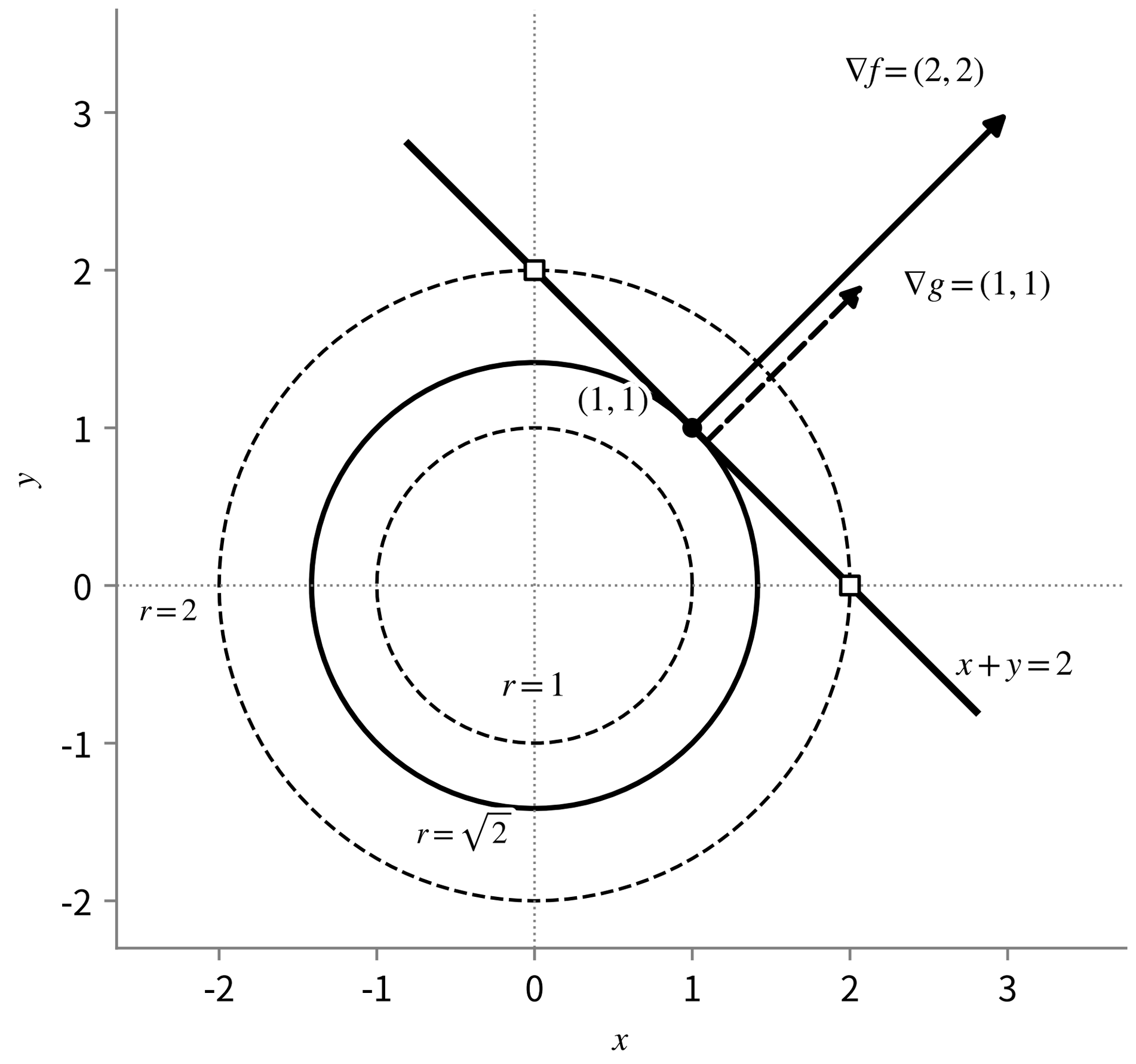

$r = 1$：不相交；$r = 2$：相交于两点（方块）